

Recitation 3

Gaussian Random Vectors and Bayesian Models
Solutions and Teaching Notes

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Pacing guide (75 minutes)

Time	Activity	Suggested use
0–25 min	Review	Gaussian vectors, mean MLE, Bayesian updating
25–37 min	Problem 1	6 min individual work, then discussion
37–55 min	Problem 2	9 min individual work, then discussion
55–69 min	Problem 3	7 min individual work, then discussion
69–75 min	Wrap-up or Problem 4	Use Problem 4 only if the class is ahead

Teaching emphasis. For Gaussian maximum likelihood, derive only the mean-vector estimate. Treat the covariance matrix as fixed during that derivation; the matrix derivative for estimating the covariance is outside today's scope. In the Bayesian section, keep the parameter random and distinguish the prior, likelihood, and posterior at every step.

1 Gaussian random vectors

A d -dimensional Gaussian random vector with mean $\boldsymbol{\mu} \in \mathbb{R}^d$ and positive-definite covariance matrix $\boldsymbol{\Sigma} \in \mathbb{R}^{d \times d}$ is written

$$\tilde{\mathbf{x}} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}).$$

Its joint pdf is

$$f_{\boldsymbol{\mu}, \boldsymbol{\Sigma}}(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^d \det(\boldsymbol{\Sigma})}} \exp\left[-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right].$$

The diagonal entry $\boldsymbol{\Sigma}_{jj}$ is the variance of coordinate j , and $\boldsymbol{\Sigma}_{jk}$ is the covariance between coordinates j and k .

Every linear transformation of a Gaussian random vector is Gaussian:

$$\mathbf{A}\tilde{\mathbf{x}} + \mathbf{b} \sim \mathcal{N}(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top).$$

Why? One characterization of a Gaussian random vector is that every linear combination of its coordinates is a univariate Gaussian random variable. Set $\tilde{\mathbf{z}} := \mathbf{A}\tilde{\mathbf{x}} + \mathbf{b}$. For any fixed vector $\mathbf{c}$,

$$\mathbf{c}^\top \tilde{\mathbf{z}} = (\mathbf{A}^\top \mathbf{c})^\top \tilde{\mathbf{x}} + \mathbf{c}^\top \mathbf{b}.$$

The right-hand side is a linear combination of the coordinates of $\tilde{\mathbf{x}}$ plus a constant, so it is Gaussian. Since this holds for every $\mathbf{c}$, $\tilde{\mathbf{z}}$ is a Gaussian random vector. Its parameters follow from linearity of the mean and the covariance rules:

$$\begin{aligned}\mathbb{E}[\tilde{\mathbf{z}}] &= \mathbf{A}\mathbb{E}[\tilde{\mathbf{x}}] + \mathbf{b} = \mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \\ \text{Cov}(\tilde{\mathbf{z}}) &= \mathbf{A} \text{Cov}(\tilde{\mathbf{x}}) \mathbf{A}^\top = \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top.\end{aligned}$$

Consequently, every subvector is Gaussian. If

$$\begin{bmatrix} \tilde{\mathbf{x}} \\ \tilde{\mathbf{y}} \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \boldsymbol{\mu}_x \\ \boldsymbol{\mu}_y \end{bmatrix}, \begin{bmatrix} \boldsymbol{\Sigma}_{xx} & \boldsymbol{\Sigma}_{xy} \\ \boldsymbol{\Sigma}_{yx} & \boldsymbol{\Sigma}_{yy} \end{bmatrix}\right),$$

then conditioning also preserves Gaussianity:

$$\tilde{\mathbf{y}} \mid \tilde{\mathbf{x}} = \mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}_y + \boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}(\mathbf{x} - \boldsymbol{\mu}_x), \boldsymbol{\Sigma}_{yy} - \boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}\boldsymbol{\Sigma}_{xy}).$$

Why? Define

$$\mathbf{B} := \boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}, \quad \tilde{\mathbf{r}} := \tilde{\mathbf{y}} - \boldsymbol{\mu}_y - \mathbf{B}(\tilde{\mathbf{x}} - \boldsymbol{\mu}_x).$$

The pair $(\tilde{\mathbf{x}}, \tilde{\mathbf{r}})$ is jointly Gaussian because it is a linear transformation of $(\tilde{\mathbf{x}}, \tilde{\mathbf{y}})$. Moreover,

$$\text{Cov}(\tilde{\mathbf{r}}, \tilde{\mathbf{x}}) = \boldsymbol{\Sigma}_{yx} - \mathbf{B}\boldsymbol{\Sigma}_{xx} = \mathbf{0}.$$

For jointly Gaussian vectors, zero covariance implies independence, so $\tilde{\mathbf{r}}$ is independent of $\tilde{\mathbf{x}}$. Its mean is zero and its covariance is

$$\text{Cov}(\tilde{\mathbf{r}}) = \boldsymbol{\Sigma}_{yy} - \boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}\boldsymbol{\Sigma}_{xy}.$$

Finally, rewrite $\tilde{\mathbf{y}}$ as

$$\tilde{\mathbf{y}} = \boldsymbol{\mu}_y + \mathbf{B}(\tilde{\mathbf{x}} - \boldsymbol{\mu}_x) + \tilde{\mathbf{r}}.$$

After conditioning on $\tilde{\mathbf{x}} = \mathbf{x}$, the first two terms are fixed and the independent Gaussian residual $\tilde{\mathbf{r}}$ is unchanged. This gives exactly the conditional mean and covariance above.

2 Maximum likelihood for a Gaussian mean vector

Suppose

$$\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}),$$

where $\boldsymbol{\Sigma}$ is fixed and positive definite. For the observed dataset $X = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$, the terms of the log-likelihood that depend on $\boldsymbol{\mu}$ are

$$\ell(\boldsymbol{\mu}) = C - \frac{1}{2} \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \boldsymbol{\mu}).$$

The gradient and Hessian are

$$\nabla_{\boldsymbol{\mu}} \ell(\boldsymbol{\mu}) = \boldsymbol{\Sigma}^{-1} \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\mu}), \quad \nabla_{\boldsymbol{\mu}}^2 \ell(\boldsymbol{\mu}) = -n\boldsymbol{\Sigma}^{-1}.$$

Because $\boldsymbol{\Sigma}^{-1}$ is positive definite, the Hessian is negative definite, so the unique maximizer is

$$\boxed{\boldsymbol{\mu}_{\text{ML}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i.}$$

Thus the ML estimate is obtained by averaging each coordinate.

3 Bayesian parametric models

In a Bayesian model, the parameter is itself random. Let $\tilde{\theta}$ be a parameter and let $\tilde{\mathbf{x}}$ represent the data. The model specifies

$$\begin{aligned} \text{prior:} & f_{\tilde{\theta}}(\theta), \\ \text{likelihood:} & f_{\tilde{\mathbf{x}}|\tilde{\theta}}(\mathbf{x} | \theta), \\ \text{posterior:} & f_{\tilde{\theta}|\tilde{\mathbf{x}}}(\theta | \mathbf{x}) = \frac{f_{\tilde{\mathbf{x}}|\tilde{\theta}}(\mathbf{x} | \theta) f_{\tilde{\theta}}(\theta)}{f_{\tilde{\mathbf{x}}}(\mathbf{x})}. \end{aligned}$$

The denominator is a normalizing constant, so as a function of θ ,

$$\boxed{\text{posterior} \propto \text{likelihood} \times \text{prior}.}$$

Two conjugate updates will be useful today. A prior is called **conjugate** when the posterior belongs to the same distribution family as the prior.

Gaussian prior and Gaussian likelihood. Suppose

$$\tilde{\mu} \sim \mathcal{N}(m_0, \tau^2), \quad \tilde{y} | \tilde{\mu} = \mu \sim \mathcal{N}(\mu, s^2),$$

where τ^2 describes prior uncertainty and s^2 describes sensor noise. Bayes' rule gives, as a function of μ ,

$$\begin{aligned} f_{\tilde{\mu}|\tilde{y}}(\mu | y) & \propto f_{\tilde{y}|\tilde{\mu}}(y | \mu) f_{\tilde{\mu}}(\mu) \\ & \propto \exp\left[-\frac{(y - \mu)^2}{2s^2}\right] \exp\left[-\frac{(\mu - m_0)^2}{2\tau^2}\right]. \end{aligned}$$

Expanding the two squares and keeping the terms that involve μ gives

$$\log f_{\tilde{\mu}|\tilde{y}}(\mu | y) = -\frac{1}{2} \left[\left(\frac{1}{\tau^2} + \frac{1}{s^2} \right) \mu^2 - 2 \left(\frac{m_0}{\tau^2} + \frac{y}{s^2} \right) \mu \right] + C.$$

Define

$$v_1 := \left(\frac{1}{\tau^2} + \frac{1}{s^2} \right)^{-1}, \quad m_1 := v_1 \left(\frac{m_0}{\tau^2} + \frac{y}{s^2} \right).$$

Completing the square turns the preceding expression into

$$-\frac{(\mu - m_1)^2}{2v_1} + C',$$

which is the log-pdf of a Gaussian distribution. Therefore

$$\boxed{\tilde{\mu} | \tilde{y} = y \sim \mathcal{N}(m_1, v_1).}$$

The quantities $1/\tau^2$ and $1/s^2$ are called **precisions**. The posterior precision is their sum, and

$$m_1 = \frac{s^2}{s^2 + \tau^2} m_0 + \frac{\tau^2}{s^2 + \tau^2} y$$

is a weighted average of the prior mean and the observation. A more precise source receives more weight. Also $v_1 < \tau^2$ and $v_1 < s^2$, so combining the prior with the observation reduces uncertainty.

Beta prior and Bernoulli likelihood. Suppose $\tilde{\theta} \sim \text{Beta}(a, b)$, so on $0 < \theta < 1$ its prior density has the form

$$f_{\tilde{\theta}}(\theta) \propto \theta^{a-1}(1-\theta)^{b-1}.$$

Given $\tilde{\theta} = \theta$, let the Bernoulli observations be conditionally independent. If the observed data contain h heads and t tails, their likelihood is

$$p_{\tilde{\theta}}(\mathbf{x} \mid \theta) = \prod_i \theta^{x_i}(1-\theta)^{1-x_i} = \theta^h(1-\theta)^t.$$

Multiplying likelihood and prior gives

$$\begin{aligned} f_{\tilde{\theta}|\mathbf{x}}(\theta \mid \mathbf{x}) &\propto \theta^h(1-\theta)^t \theta^{a-1}(1-\theta)^{b-1} \\ &= \theta^{a+h-1}(1-\theta)^{b+t-1}. \end{aligned}$$

This is the kernel of another beta density, so

$$\boxed{\tilde{\theta} \mid \text{data} \sim \text{Beta}(a+h, b+t).}$$

The update simply adds the observed counts to the two beta parameters. For example, the uniform prior is $\text{Beta}(1, 1)$; after two heads it becomes $\text{Beta}(3, 1)$.

Practice problems

Problems 2 and 3 are adapted from Exercises 5.15 and 6.14 of *Probability and Statistics for Data Science*.

Problem 1: MLE of a Gaussian mean vector

[6 minutes]

Let

$$\tilde{\mathbf{x}}_1, \tilde{\mathbf{x}}_2, \tilde{\mathbf{x}}_3 \stackrel{\text{iid}}{\sim} \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}), \quad \boldsymbol{\Sigma} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix},$$

and suppose the observed vectors are

$$\mathbf{x}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 4 \\ 0 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

- (a) Write the log-likelihood as a function of $\boldsymbol{\mu}$, up to an additive constant.
- (b) Differentiate the general n -observation expression and derive $\boldsymbol{\mu}_{\text{ML}}$.
- (c) Compute $\boldsymbol{\mu}_{\text{ML}}$ for the observed dataset.
- (d) Would replacing $\boldsymbol{\Sigma}$ by the identity matrix change this ML estimate? Briefly explain.

Solution. For these three observations,

$$\ell(\boldsymbol{\mu}) = C - \frac{1}{2} \sum_{i=1}^3 (\mathbf{x}_i - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \boldsymbol{\mu}).$$

For general n ,

$$\nabla_{\boldsymbol{\mu}} \ell(\boldsymbol{\mu}) = \boldsymbol{\Sigma}^{-1} \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\mu}).$$

Since $\boldsymbol{\Sigma}$ is invertible, setting the gradient to zero gives

$$\sum_{i=1}^n \mathbf{x}_i - n\boldsymbol{\mu} = \mathbf{0} \quad \implies \quad \boldsymbol{\mu}_{\text{ML}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i.$$

The negative-definite Hessian $-n\boldsymbol{\Sigma}^{-1}$ confirms that this is the unique maximum. For the observed data,

$$\boldsymbol{\mu}_{\text{ML}} = \frac{1}{3} \left(\begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 4 \\ 0 \end{bmatrix} + \begin{bmatrix} 3 \\ 2 \end{bmatrix} \right) = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

Replacing the common covariance by the identity changes the likelihood contours but not the maximizing mean. The same factor $\boldsymbol{\Sigma}^{-1}$ multiplies every residual and cancels from the score equation.

Problem 2: Gaussian Bayesian model (Exercise 5.15)**[9 minutes]**

A noisy sensor is used to estimate the temperature on top of a mountain. Let $\tilde{\mu}$ denote the unknown temperature. The prior and sensor model are

$$\tilde{\mu} \sim \mathcal{N}(0, \sigma^2), \quad \tilde{y} \mid \tilde{\mu} = \mu \sim \mathcal{N}(\mu, 1).$$

- (a) Show that $[\tilde{\mu}, \tilde{y}]^\top$ is a Gaussian random vector with mean zero, and derive its covariance matrix.
- (b) Derive the posterior distribution of $\tilde{\mu}$ given $\tilde{y} = y$.
- (c) Let $y = -1$ and $\sigma^2 = 0.01$. Describe or sketch the prior and posterior. Does the prior express confidence that the temperature is close to 0? Does one reading change it much?
- (d) Repeat part (c) for $y = -1$ and $\sigma^2 = 10$.

Solution. Write $\tilde{y} = \tilde{\mu} + \tilde{\varepsilon}$, where $\tilde{\varepsilon} \sim \mathcal{N}(0, 1)$ is independent of $\tilde{\mu}$. Then

$$\begin{bmatrix} \tilde{\mu} \\ \tilde{y} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \tilde{\mu} \\ \tilde{\varepsilon} \end{bmatrix},$$

so it is a linear transformation of a Gaussian random vector and is therefore Gaussian. Its mean is zero and its covariance matrix is

$$\begin{bmatrix} \text{Var}(\tilde{\mu}) & \text{Cov}(\tilde{\mu}, \tilde{y}) \\ \text{Cov}(\tilde{y}, \tilde{\mu}) & \text{Var}(\tilde{y}) \end{bmatrix} = \begin{bmatrix} \sigma^2 & \sigma^2 \\ \sigma^2 & \sigma^2 + 1 \end{bmatrix}.$$

Using the conditional-Gaussian formula,

$$\tilde{\mu} \mid \tilde{y} = y \sim \mathcal{N}\left(\frac{\sigma^2}{\sigma^2 + 1}y, \frac{\sigma^2}{\sigma^2 + 1}\right).$$

The posterior mean is the sensor reading shrunk toward the prior mean 0.

For $y = -1$ and $\sigma^2 = 0.01$, the prior is $\mathcal{N}(0, 0.01)$ and the posterior is

$$\mathcal{N}\left(-\frac{1}{101}, \frac{1}{101}\right) \approx \mathcal{N}(-0.0099, 0.00990).$$

The prior standard deviation is 0.1, and the posterior standard deviation is about 0.0995. The two narrow curves almost overlap. This prior expresses strong confidence near 0, so one noisy reading barely changes it.

For $y = -1$ and $\sigma^2 = 10$, the prior is $\mathcal{N}(0, 10)$ and the posterior is

$$\mathcal{N}\left(-\frac{10}{11}, \frac{10}{11}\right) \approx \mathcal{N}(-0.909, 0.909).$$

The prior standard deviation is about 3.16, so it does not express strong confidence near 0. The posterior standard deviation is about 0.953 and its center is close to the reading, so the posterior is much more concentrated and meaningfully different from the prior.

Problem 3: Two coin flips (Exercise 6.14)**[7 minutes]**

Let the probability of heads be a random variable $\tilde{\theta} \sim \text{Uniform}[0, 1]$. Given $\tilde{\theta} = \theta$, the two flips $\tilde{x}_1$ and $\tilde{x}_2$ are conditionally independent Bernoulli random variables with parameter θ , where 1 denotes heads.

- (a) Compute the joint pmf of $\tilde{x}_1$ and $\tilde{x}_2$.
- (b) Both flips are observed to be heads. Compute $\mathbb{P}(\tilde{\theta} < 1/2 \mid \tilde{x}_1 = 1, \tilde{x}_2 = 1)$.
- (c) Are the two flips independent before conditioning on $\tilde{\theta}$? Use your answer to part (a) to justify your conclusion.

Solution. For $a, b \in \{0, 1\}$, integrate out the shared random parameter:

$$p_{\tilde{x}_1, \tilde{x}_2}(a, b) = \int_0^1 \theta^{a+b} (1 - \theta)^{2-a-b} d\theta.$$

Therefore

	$\tilde{x}_2 = 0$	$\tilde{x}_2 = 1$
$\tilde{x}_1 = 0$	$\frac{1}{3}$	$\frac{1}{6}$
$\tilde{x}_1 = 1$	$\frac{1}{6}$	$\frac{1}{3}$

The posterior density after two heads is proportional to θ^2 on $[0, 1]$. Since $\int_0^1 \theta^2 d\theta = 1/3$,

$$f_{\tilde{\theta}|\tilde{x}_1, \tilde{x}_2}(\theta \mid 1, 1) = 3\theta^2, \quad 0 \leq \theta \leq 1.$$

Hence

$$\mathbb{P}(\tilde{\theta} < 1/2 \mid \tilde{x}_1 = 1, \tilde{x}_2 = 1) = \int_0^{1/2} 3\theta^2 d\theta = \boxed{\frac{1}{8}}.$$

Each flip has marginal probability $1/2$ of heads, but

$$\mathbb{P}(\tilde{x}_1 = 1, \tilde{x}_2 = 1) = \frac{1}{3} \neq \frac{1}{2} \cdot \frac{1}{2}.$$

Thus the flips are not marginally independent. They are independent only after conditioning on their shared parameter $\tilde{\theta}$.

Problem 4 (backup): True or false?

Optional if time allows. Suggested working time: 5–7 minutes.

- (a) For a jointly Gaussian vector, two coordinates with covariance zero are independent.
- (b) If every observation has the same known covariance matrix, changing that matrix changes the ML estimate of the Gaussian mean vector.
- (c) In a Bayesian model, the posterior is proportional to the prior divided by the likelihood.
- (d) Random variables may be conditionally independent given a shared parameter but dependent after that parameter is integrated out.

Solution. The answers are **true, false, false, true**.

- (a) This implication is a special property of jointly Gaussian random vectors; it is not true for arbitrary distributions.
- (b) With a common positive-definite covariance, the score equation reduces to $\sum_i (\mathbf{x}_i - \boldsymbol{\mu}) = \mathbf{0}$, so the estimate remains the coordinatewise sample mean.
- (c) The posterior is proportional to likelihood times prior.
- (d) Exercise 6.14 is an example: the flips are conditionally independent given $\tilde{\theta}$ but positively dependent marginally.